Skill-3D: Evolving Scene-Aware Skills for Agentic 3D Spatial Reasoning

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HarnessRSI 63 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

Skill = trigger, evidence, tool order / arguments, evidence-to-answer rule.

1. Scene + task → tool workflow
Retrieve skills, select a compact set, execute perception / geometry tools.

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2. Record outcome in Scene Memory
Store visual context, tool evidence, answer and outcome. Success-label procedure is not detailed.

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3. Success → reusable procedure
New skill or useful-coverage merge. Redundant success changes statistics only.

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4. Failure → lesson / fallback
Attach supported lessons; patch reliable fallbacks. Static seeds stay fixed.

↓

5. Skill Manager → accept or reject
Keep supported updates consistent with prior successes. No numeric or rerun gate specified.

Repeat on pooled training questions.
GPT-5.4 teacher; freeze library for testing.
Qwen additionally uses SFT + GRPO.

2 Compare against tool agents too

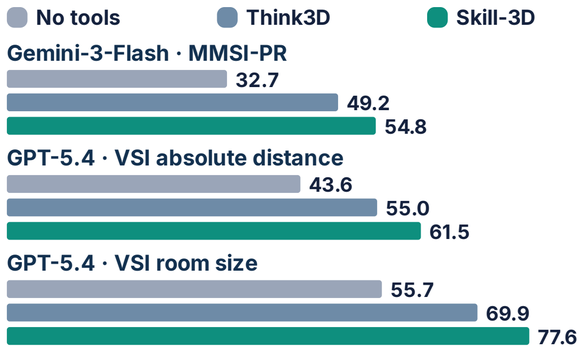


Table 1 · question-held-out test. MMSI 67% is vs no tools. No main-table run counts, variance or CI.

4 Where gains are small / uncertain

- **CV-3D · Gemini-2.5-Pro:** depth order 94.0 vs Think3D 92.7; relative distance 92.8 vs 91.6. No CI.
- **Qwen3-VL-8B · post-trained:** MMSI-PR 42.8 vs Think3D-8B 41.2; VSI room size 67.9 vs 58.1.
- **GPT-5.4 inference · Table B.1:** Skill-3D 2.6 calls / 20.8 s vs Think3D 1.8 / 35.1 s. More calls, less time; direct tools are faster (13.2 s).
- **Indoor visual QA, not simulator exploration.** Question holdout does not rule out scene/video overlap.

3 Controls—and their boundaries

GPT-5.4 ablations · Table 3



Reported Avg.; aggregate scope unresolved (SOURCE.md). No variance.

Transfer · fixed static skills, GPT-5.4

Target / dynamic source	none	VSI
MMSI positional relation	52.6	**57.8**
CV-3D evaluated-task Avg.	92.2	**93.7**

B.2: shared static priors; no CI. No-dynamic VSI: 67.8 (T3) vs 63.4 (B.2); scope/config unexplained. Compare within tables.

What the controls do not establish

- Question holdout; no scene-overlap audit. BLINK: multi-view subset.
- No tools / direct tools / Think3D: no reported equal-library or matched construction-budget comparison.
- Tuning split / baseline tuning budgets unstated; k, tool budget B, numeric gate unspecified.
- 3 h SFT + 28 h RL: post-training, not evolution cost. Teacher/construction totals unreported.

Takeaway

In Table 3's reported ablations, dynamic skills help, but static priors and retrieval matter more. Transfer exists. Equal-budget skill learning and scene-held-out generalization remain unverified.

Our read · HarnessRSI

63 · ★★ · evidence: retained
Reports: holdout, ablations, transfer.
Missing: evolution cost, matched budgets, main-score uncertainty, scene audit.